1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 highlights that mismatch-repair status (MSI/dMMR) and specific MMR gene alteration (MSH6) should increase neoantigen load and TIL infiltration in NMIBC, yet immunotherapy/BCG benefit is inconsistent — why do abundant TILs in some MMR-deficient tumors fail to translate into effective local therapy response? 2) Tier 1 paths used: (a) MSI/dMMR path — motivates the neoantigen/TIL axis and immunotherapy relevance across cancers; (b) MSH6 prognostic path — provides a concrete MMR gene whose loss can drive MSI and shape tumor immunogenicity; I built the hypothesis on the tension between MMR-driven immunogenicity and variable clinical response. 3) Tier 2 support: BCG/immune-activation path supplies the clinical therapy context (intravesical immune stimulation) whose efficacy might be modulated by tumor MMR state; liquid-biopsy ADNP and systemic metabolic markers (HDL-C, ALP, IGF1) provide plausible non■tumor or systemic modifiers of local immune competence and could explain residual heterogeneity in TIL function or therapy outcome. 4) Endpoint choice: I chose therapy_response (BCG) because the core question is why immune engagement (TILs driven by MMR deficiency) does not reliably predict benefit from a local immune-based therapy in NMIBC — this is the most actionable clinical readout for NMIBC management. 5) Deliberate exclusions: I excluded exploratory/tangential paths (HDL-C mechanistic details, PSA cross-disease uses, CSF AD biomarkers) from the core claim to keep the hypothesis tightly focused on the MMR–TIL–therapy interaction; those items are reserved as secondary_modifiers for follow-up analyses?

Working Hypothesis

In NMIBC, MSH6 loss and MSI/dMMR drive elevated neoantigen burden and high TIL density, but promote an exhausted T cell phenotype and myeloid-suppressive microenvironment that impairs local anti-tumor immunity. Therefore, among NMIBC tumors with high TIL density, those with MSH6-deficient/MSI-high status will have poorer response to intravesical BCG than tumors that are MSH6-intact/MSI-stable, establishing a predictive interaction between TIL density, MSH6/MSI status, and BCG therapy outcome.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, MSI/dMMR
Primary endpoint	therapy_response (BCG)

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6) should increase neoantigen load and TIL infiltration in NMIBC, yet immunotherapy/BCG benefit is inconsistent — why do abundant TILs in some MMR-deficient tumors fail to translate into effective local therapy response? 2) Tier 1 paths used: (a) MSI/dMMR path — motivates the neoantigen/TIL axis and immunotherapy relevance across cancers; (b) MSH6 prognostic path — provides a concrete MMR gene whose loss can drive MSI and shape tumor immunogenicity; I built the hypothesis on the tension between MMR-driven immunogenicity and variable clinical response.

MSI/dMMR: MSI/dMMR) and specific MMR gene alteration (MSH6) should increase neoantigen load and TIL infiltration in NMIBC, yet immunotherapy/BCG benefit is inconsistent — why do abundant TILs in some MMR-deficient tumors fail to translate into effective local therapy response? 2) Tier 1 paths used: (a) MSI/dMMR path — motivates the neoantigen/TIL axis and immunotherapy relevance across cancers; (b) MSH6 prognostic path — provides a concrete MMR gene whose loss can drive MSI and shape tumor immunogenicity; I built the hypothesis on the tension between MMR-driven immunogenicity and variable clinical response.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	ADNP (liquid biopsy transcript)	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	IGF1	Supporting modifier from knowledge graph
Secondary	BCG_vaccine (therapy context)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective and prospective NMIBC cohorts with available pathology (IHC for MSH6, CD3/CD8; MSI testing) and documented BCG outcomes.

Prediction 2: Multi-omics and immunophenotyping (PD-1/PD-L1, LAG-3, myeloid markers, RNA panels including ADNP, IGF1; serum HDL-C/ALP as systemic modifiers).

Prediction 3: Functional assays with patient-derived TILs to assess exhaustion and cytotoxic capacity; optional in vivo models (Msh6-deficient bladder cancer in mice with BCG challenge) to validate causality. Analysis: test the interaction between TIL density and MSH6/MSI status on BCG response using logistic regression with an interaction term; adjust for confounders (tumor stage, prior therapy, age, sex).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ADNP (liquid biopsy transcript): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.